

Vanilla RNN

x_t, h_{t-1}

overwrite

one hidden state
 $h_t = \tanh(\dots)$

output uses h_t

GRU

x_t, h_{t-1}

reset + update gates

blend old h
with candidate

h_t is also output state

LSTM

x_t, h_{t-1}, c_{t-1}

forget + input + output gates

protected cell memory c_t

$o_t * \tanh(c_t)$

exposed hidden state h_t